

MAT 426 - Topics in Linear Algebra – Exam 2 Review Exercises

Below are some questions to help you prepare for Exam 2 on Tuesday, April 28. This list is not meant to be exhaustive and some topics may have been missed. Any topics that I have failed to include here are still considered reasonable exam material. Be sure to study all materials considered in class and in the homework.

1. Prove the Orthogonal Decomposition Theorem: Let W be a subspace of $\mathbb{R}^n$. Then each $\mathbf{y}$ in $\mathbb{R}^n$ can be written uniquely in the form $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$ where $\hat{\mathbf{y}}$ is in W and $\mathbf{z}$ is in $W^\perp$. If $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ is any orthogonal basis of W then

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \cdots + \frac{\mathbf{y} \cdot \mathbf{u}_p}{\mathbf{u}_p \cdot \mathbf{u}_p} \mathbf{u}_p$$

and $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$.

2. Prove the Best Approximation Theorem: Let W be a subspace of $\mathbb{R}^n$, let $\mathbf{y}$ be any vector in $\mathbb{R}^n$, and let $\hat{\mathbf{y}}$ be the orthogonal projection of $\mathbf{y}$ onto W . Then, $\hat{\mathbf{y}}$ is the closest point in W to $\mathbf{y}$, in the sense that

$$\|\mathbf{y} - \hat{\mathbf{y}}\| < \|\mathbf{y} - \mathbf{v}\|$$

for all $\mathbf{v}$ in W distinct from $\hat{\mathbf{y}}$.

3. Let $W = \text{Span} \left\{ \mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} -1 \\ 3 \\ -2 \end{bmatrix} \right\}$. Write $\mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ as the sum of a vector in W and a vector orthogonal to W .

4. Let $W = \text{Span} \left\{ \mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \mathbf{u}_3 = \begin{bmatrix} 0 \\ -1 \\ 1 \\ -1 \end{bmatrix} \right\}$. Write $\mathbf{y} = \begin{bmatrix} 3 \\ 4 \\ 5 \\ 4 \end{bmatrix}$ as the sum of a vector in W and a vector orthogonal to W .

5. Find the best approximation to $\mathbf{z} = \begin{bmatrix} 2 \\ 4 \\ 0 \\ -1 \end{bmatrix}$ by vectors of the form $c_1 \begin{bmatrix} 2 \\ 0 \\ -1 \\ -3 \end{bmatrix} + c_2 \begin{bmatrix} 5 \\ -2 \\ 4 \\ 2 \end{bmatrix}$.

6. The given set is a basis for a subspace W . Use the Gram-Schmidt process to produce an orthogonal basis for W .

$$(a) \left\{ \begin{bmatrix} 3 \\ 1 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} -5 \\ 1 \\ 5 \\ -7 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -2 \\ 8 \end{bmatrix} \right\} \qquad (b) \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ -3 \\ 2 \\ 5 \\ 5 \end{bmatrix}, \begin{bmatrix} 5 \\ 1 \\ 3 \\ 2 \\ 8 \end{bmatrix} \right\}$$

7. Find a QR decomposition for the following matrices.

$$(a) A = \begin{bmatrix} 2 & 8 & 2 \\ 1 & 7 & -1 \\ -2 & -2 & 1 \end{bmatrix} \qquad (b) A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \\ -1 & -1 \\ 0 & 1 \end{bmatrix} \qquad (c) A = \begin{bmatrix} 1 & 2 & 2 \\ -1 & 1 & 2 \\ -1 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

8. Outline the QR Iteration Algorithm to compute the eigenvalues of a matrix A .

9. State the definition of a Householder matrix H .
10. Let H be a Householder matrix with nonzero vector $\mathbf{u} \in \mathbb{R}^n$. Prove each of the following.
- (a) $\|H\mathbf{x}\| = \|\mathbf{x}\|$ for any $\mathbf{x} \in \mathbb{R}^n$
 - (b) $H^2 = I$
 - (c) $H\mathbf{u} = -\mathbf{u}$
 - (d) $H\mathbf{v} = \mathbf{v}$ if $\mathbf{v}^T\mathbf{u} = 0$
11. Determine the eigenvalues of a Householder matrix H and their algebraic multiplicities. Use this to determine the determinant of H .
12. For the given nonzero vector $\mathbf{u}$, find the Householder matrix H .

$$(a) \mathbf{u} = \begin{bmatrix} -3 \\ 4 \end{bmatrix} \quad (b) \mathbf{u} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} \quad (c) \mathbf{u} = \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix}$$

13. For each of the vectors $\mathbf{x}$ below, create a Householder matrix H so that $H\mathbf{x}$ is a multiple of $\mathbf{e}_1$.

$$(a) \mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \quad (b) \mathbf{x} = \begin{bmatrix} -1 \\ 0 \\ 1 \\ -1 \end{bmatrix} \quad (c) \mathbf{x} = \begin{bmatrix} 4 \\ 3 \\ -2 \\ 1 \end{bmatrix}$$

14. For each of the following matrices A find the QR factorization of A using Householder matrices.

$$(a) A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \\ 2 & -2 & 4 \end{bmatrix} \quad (b) A = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 1 & 3 \\ 2 & 1 & 2 \\ 0 & 3 & -2 \end{bmatrix}$$

15. What are dimension of the matrices in the QR factorization using Gram-Schmidt? What are the dimensions of the matrices in the QR factorization using Householder matrices? Do you obtain a full QR factorization from a reduced QR factorization?
16. Define what it means for a matrix A to be orthogonally diagonalizable.
17. Prove that if A is orthgonally diagonalizable, then A is symmetric.
18. Prove that the eigenvalues of a real symmetric matrix are real.
19. Prove that if A is symmetric, then any two eigenvectors corresponding to distinct eigenvalues are orthogonal.
20. Orthogonally diagonalize the matrix A by finding an orthogonal matrix Q and a diagonal matrix D such that $Q^T A Q = D$. Also, find the spectral decomposition of A .

$$(a) A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \quad (b) A = \begin{bmatrix} 2 & 3 & 0 \\ 3 & 2 & 4 \\ 0 & 4 & 2 \end{bmatrix} \quad (c) A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \quad (d) A = \begin{bmatrix} 2 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 2 \end{bmatrix}$$

21. If A and B are orthogonally diagonalizable and $AB = BA$, show that AB is orthogonally diagonalizable.

22. Suppose $A = PRP^{-1}$ where P is orthogonal and R is upper triangular. Show that if A is symmetric, then R is symmetric and hence is a diagonal matrix.
23. Find a 3×3 matrix with eigenvalues $\lambda_1 = 1$, $\lambda_2 = 2$, and $\lambda_3 = 3$ and corresponding eigenvectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ -0 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$, and $\mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$.
24. Determine the quadratic form $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ for the given matrix A . Also, find an orthogonal matrix P such that the change of variables $\mathbf{x} = P\mathbf{y}$ transforms the quadratic form into one with no cross-product terms; state this new quadratic form.
- (a) $A = \begin{bmatrix} 3 & -2 \\ -2 & 4 \end{bmatrix}$ (b) $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ (c) $A = \begin{bmatrix} 3 & 1 & 5 \\ 1 & 4 & -5 \\ 5 & -5 & -3 \end{bmatrix}$
25. Find the symmetric matrix A associated with the given quadratic form.
- (a) $Q(\mathbf{x}) = -2x_1^2 + 5x_2^2 - x_1x_2$
 (b) $Q(\mathbf{x}) = 7x^2 + 5xy$
 (c) $Q(\mathbf{x}) = x^2 + 7y^2 - 9z^2 + xz - 2xy - 9yz$
26. Prove the Principal Axes Theorem: Let A be an $n \times n$ symmetric matrix. Then, there exists an orthogonal change of variable, $\mathbf{x} = P\mathbf{y}$ that transforms the quadratic form $\mathbf{x}^T A \mathbf{x}$ into a quadratic form $\mathbf{y}^T D \mathbf{y}$ with no cross-product terms.
27. Prove the following: Let $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ be a quadratic form with associated $n \times n$ symmetric matrix A . Let the eigenvalues of A be $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$. Then the following are true, subject to the constraint $\|\mathbf{x}\| = 1$
- (a) $\lambda_1 \geq Q(\mathbf{x}) \geq \lambda_n$
 (b) The maximum value of $Q(\mathbf{x})$ is λ_1 and it occurs when $\mathbf{x}$ is a unit eigenvector corresponding to λ_1 .
 (c) The minimum value of $Q(\mathbf{x})$ is λ_n and it occurs when $\mathbf{x}$ is a unit eigenvector corresponding to λ_n .
28. Prove the Spectral Theorem.
29. Let A be an $n \times n$ matrix, all of whose eigenvalues are real. Adapt the proof of the Spectral Theorem to show that there exist an orthogonal matrix Q and an upper triangular matrix T such that $Q^T A Q = T$.
30. Find the singular value decomposition of the matrix A below.

(a) $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ (b) $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 2 & -2 \end{bmatrix}$ (c) $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$